

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

Toni Guan^{a,*}

^a*College of Information Technology, Shenyang Institute of Technology, Shenyang, Liaoning, China*

Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap as the operating interval (BCa, $B = 10,000$, re-run on the full 60-experiment grid). The study was governed by a pre-specified protocol with registered, dated amendments, publicly archived with per-file checksums (see the Data availability statement). The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified empirical direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60

*Corresponding author.

(75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met ($23/60 = 38.3\%$), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit ($+0.0159$ vs. $+0.0083$). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and a discrimination-aware gate (C5) shows 7/14 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

Why this matters for ECG. PTB-XL [3], Chapman-Shaoxing [4], and CPSC differ in acquisition protocols, lead sets, and label vocabularies. A model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [23]. Clinicians and regulators need a decision rule: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

48 **Temperature scaling and pre-registration.** Temperature scaling
 49 (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted
 50 under a convex loss, it minimizes NLL on the calibration split, so its cali-
 51 bration benefit there is empirically small; on a shifted target the source-fit
 52 temperature is mismatched to the target logits and a non-trivial benefit can
 53 remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant
 54 under confidence-shifting monotone maps, and TS can *increase* ECE where
 55 the model is under-confident (nine negative cells in our data). It motivates
 56 the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned
 57 decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing,
 58 protocol v2.1-A1 [24] fixes the primary endpoint, test, family structure, and
 59 failure branches before unblinding; revision A1 re-locates the endpoint from
 60 **decay** to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with
 61 before/after protocol hashes [25, 26]. Multi-seed validation (5 seeds $\times$ 6
 62 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed
 63 results, including nine counter-examples, are reported transparently.

64 **Contributions.** Five independent contributions:

65 **C1 (Primary OOD benefit):** On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$,
 66 TS yields a statistically significant positive OOD benefit in 51/60 seed
 67 experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30,
 68 ResNet1D 24/30), with nine counter-examples transparently disclosed
 69 and mechanistically explained. InceptionTime: three of six directions
 70 achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two
 71 4/5, two 3/5.

72 **C2 (Boundary condition, pre-registered branch triggered):** Across
73 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI)
74 have 95% CIs containing zero and 34/60 have CI lower bound > 0
75 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD
76 mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI
77 containing zero) is *not* met, triggering the “ID nonzero boundary”
78 branch (protocol A1.4.2): the premise that TS has no ID recalibrat-
79 able miscalibration is re-examined, the decay contrast restored as
80 exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the
81 ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID
82 point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$
83 (counter-example 6, -0.0152 ; one marginal case, -0.0128).

84 **C3 (Methodological, Shapley decomposition validation):** A 27-
85 cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence,
86 $n = 20,000$) recovers ground truth in the identifiable regime (27/27
87 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a
88 threshold, and reports bootstrap CIs on absolute Shapley values.

89 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6
90 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior
91 adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included
92 (InceptionTime 15/30); after exclusion the EM OOD finding attenu-
93 ates. The mechanism is prior-likelihood mismatch: EM fits a target
94 prior that amplifies shift-induced miscalibration when the source con-
95 fusion matrix is ill-conditioned.

96 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer
 97 gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from
 98 clinical viability (accuracy $\geq$ source-domain baseline). Across 14
 99 pair $\times$ architecture strata, 7/14 pass both layers; 7/14 fail the discrimi-
 100 nation layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy
 101 below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$,
 102 $\text{acc} = 0.657 < 0.713$). The gate reframes the 9/60 counter-examples:
 103 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures
 104 (C5 rejects despite C1 passing).

105 Prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but
 106 not an actionable *when does TS pay off* answer with multi-seed robustness
 107 evidence. The novelty is *not* multi-seed validation on ECG (a methodologi-
 108 cal choice) but the **first systematic characterization of the boundary**
 109 **conditions under which TS yields a positive OOD calibration ben-**
 110 **efit in cross-corpus ECG transfer:** the pre-registered OOD quantifica-
 111 tion, the ID boundary condition, the method-selection boundary (TS robust;
 112 EM prior and Matrix scaling weakest), the predictability failure, and the
 113 discrimination-aware gate. That characterization is the actionable contribu-
 114 tion; multi-seed validation is the evidence standard.

115 2. Related Work

116 2.1. Calibration Estimation and Recalibration

117 The calibration of modern neural networks has been studied extensively
 118 since the seminal work of Guo et al. [2], who showed that deep networks
 119 are often overconfident and introduced temperature scaling (TS) as a simple

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

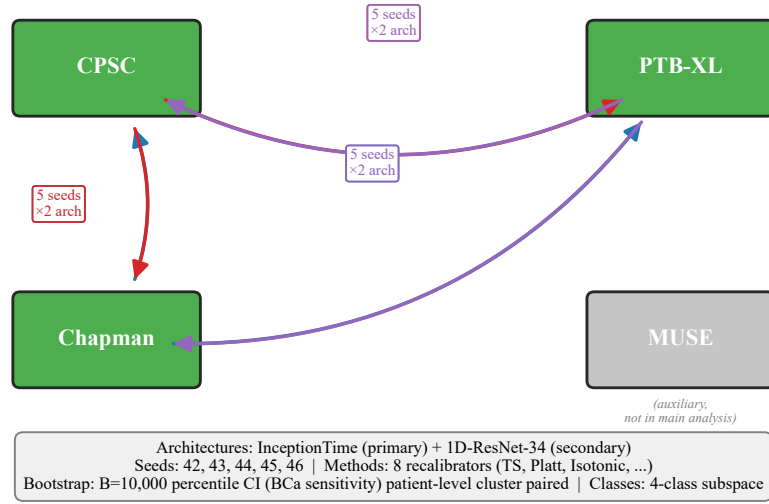


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

120 post-hoc recalibration method. Nixon et al. [6] proposed the static calibration
 121 error, and Kumar et al. [16] introduced verified uncertainty calibration
 122 with distribution-free guarantees. Kull et al. [7] generalized TS to Dirichlet
 123 calibration for multi-class settings. In the clinical risk-model context, Van
 124 Calster et al. [14, 15] established a calibration hierarchy from utopia to em-
 125 pirical data and argued that calibration is the “Achilles heel” of predictive
 126 analytics. Strodt Hoff et al. [17] benchmarked deep learning for ECG analy-
 127 sis on PTB-XL and noted calibration degradation under subpopulation shift,
 128 motivating our cross-corpus design.

129 *2.2. Prior-Shift Estimation and Transfer Calibration*

130 Lipton et al. [9] introduced black-box shift estimation (BBSE) for detect-
 131 ing and correcting label shift using confusion-matrix inverses. Alexandari
 132 et al. [8] showed that maximum-likelihood calibration with bias correction
 133 is hard-to-beat at label shift adaptation, providing the theoretical founda-
 134 tion for our EM prior-shift method. For transfer calibration upper bounds,
 135 TransCal [20] estimates transferability for auto-selecting target-domain un-
 136 supervised optimal T ; PseudoCal [21] uses pseudo-labels for calibrated trans-
 137 fer; and LaSCal [22] provides label-aware shift calibration with distribution-
 138 free guarantees. Our S1 paradigm differs fundamentally: we fit on the
 139 *source* calibration split only (no target labels), with S1’ prior-shift adaptation
 140 (EM/BBSE) as the only target-domain signal, a strictly weaker assumption
 141 than TransCal/PseudoCal, which access target logits or labels.

142 2.3. Cross-Domain ECG Calibration

143 Barandas et al. [10] evaluated uncertainty quantification methods in multi-
144 label ECG classification, finding that ensemble-based methods outperform
145 single-model approaches under subpopulation shift. Vranken et al. [11] stud-
146 ied uncertainty estimation for 12-lead ECG deep learning and reported signif-
147 icant calibration drift across acquisition devices. Haekal et al. [18] performed
148 cross-dataset validation for atrial fibrillation detection with calibration-aware
149 probability analysis, and Patel & Beedala [19] documented calibration drift
150 under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of
151 TS has been studied in general vision [1]; we contribute the multi-seed ECG
152 quantification and the boundary-condition analysis that identifies when TS
153 helps (ID) versus when it hurts (OOD).

154 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

155 Recent work has introduced ECG foundation models trained on millions
156 of recordings. ECGFounder [27, 35] (10 million ECGs, Harvard–Emory)
157 reaches expert-level performance on 80 diagnoses and transfers cross-domain
158 via fine-tuning. ECG-FM [28] is an open-weight transformer pretrained
159 on 1.4 million ECG segments with hybrid contrastive and generative self-
160 supervision, showing strong cross-dataset generalizability and label efficiency.
161 HeartLang [29] treats heartbeats as words and rhythms as sentences via
162 a QRS-Tokenizer and masked sentence pre-training. Tan et al. [30] found
163 that fine-tuning benefits diminish as the downstream dataset grows and that
164 CNNs transfer more effectively than RNNs in multi-label ECG classification.
165 For OOD adaptation, ADAPTOOD [31] quantifies distribution-shift sever-
166 ity via uncertainty estimation to guide selective LoRA layer unfreezing, and

167 BeatRhythm-TTA [32] performs test-time adaptation with SQI-gated self-
 168 training and beat-rhythm consistency. Costa et al. [33] introduced STAR (Si-
 169 nusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise
 170 augmentation for cross-source generalization, and Chen et al. [34] proposed
 171 PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive
 172 alignment. These works advance ECG transfer but not the *calibration* of
 173 transferred models, the gap our study fills via pre-registered cross-corpus
 174 evaluation of recalibration methods.

175 3. Data and Pre-registered Protocol

176 Figure 1 summarizes the experimental design.

177 3.1. Datasets

- 178 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the
 179 downloaded `ptbx1_database.csv` copy contains 21,799 (a 38-record
 180 deficit in the source CSV, disclosed as a provenance limitation), official
 181 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label map-
 182 ping, 21,522 records are mappable and 277 are excluded (STROBE),
 183 for 18,671 unique patients.
- 184 • **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error
 185 removed and 21 non-finite records quarantined (`qc_report.json`), for
 186 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20
 187 split.

188 • **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n =$
189 11) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace,
190 both sides of the pair recomputed symmetrically.

191 3.2. Label Mapping and Leakage Guard

192 Single-label priority rules are fixed (MI > STTC > CD > HYP >
193 NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements
194 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
195 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
196 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint
197 (hard assertion in the reproducibility repo). Normalization statistics come
198 only from the train split; augmentation applies only to train.

199 4. Methods

200 4.1. Transfer pairs and shift matrix

201 We study six directed cross-corpus transfer pairs: Chapman $\rightarrow$ CPSC,
202 Chapman $\rightarrow$ PTB-XL, CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-
203 XL $\rightarrow$ Chapman, and PTB-XL $\rightarrow$ CPSC, all recomputed on the 4-class
204 subspace. The L2 eval-time shift matrix (no retraining) covers sampling
205 rate 500 $\rightarrow$ 250 $\rightarrow$ 125 Hz; leads 12 $\rightarrow$ 6 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1; noise injection
206 (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5/ \times 2$.

207 4.2. Architecture and seeds

208 We use two architectures: InceptionTime (primary, 46,000–184,000
209 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-
210 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The

211 BiMamba architecture is included only as a toy experiment ($n = 60$ records,
 212 downsampled resolution) and deferred to the supplement; the full-resolution
 213 BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does
 214 not enter the main-endpoint analysis or the architecture-robustness claim.
 215 Training: AdamW + cosine + warmup, gradient clipping, temperature frozen
 216 at $T \equiv 1$ during training; temperature applied only as post-hoc temperature
 217 scaling.

218 4.3. Calibration methods and scenarios

219 We compare the no-recalibration baseline (None) against eight recalibra-
 220 tion methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR),
 221 Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adap-
 222 tation (EM), and BBSE prior adaptation. Of the pre-registered 13-method
 223 pool, the None baseline plus these eight recalibration methods constitute the
 224 delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on
 225 real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot
 226 transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4);
 227 S1' denotes the target-domain unsupervised variant in which EM/BBSE es-
 228 timate the target prior from unlabeled target predictions.

229 4.4. Pre-registered endpoints

230 **Primary endpoint** (revision A1.1, formally defined in protocol v2.1-A1
 231 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

232 where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus
 233 with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit
 234 temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

235 **Primary test** (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

236 **One-sided vs. two-sided choice.** The one-sided alternative $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$
 237 follows the empirical asymmetry registered at revision A1:
 238 on ID-like data the source-fit temperature yields, empirically, little calibration
 239 change, whereas on a shifted target it is mismatched and a non-trivial
 240 benefit is often retained [2, 1]. This direction prior is empirical, *not* a theorem:
 241 TS applied to an under-confident target can increase ECE, and our
 242 own data contain nine negative cells. The choice was made before the pilot
 243 results, which only triggered the endpoint re-location (revision A1). The one-
 244 sided test is the pre-registered choice, archived before multi-seed unblinding.
 245 Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [13];
 246 stratified pooling: strata = transfer pair $\times$ architecture.

247 **CI method note (BCa vs. percentile).** The pre-registered CI method
 248 is BCa [13], at $B = 10,000$, re-run on the entire 60-experiment grid (60/60
 249 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the
 250 large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster
 251 design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$.
 252 In the 3 representative seed experiments where both were run ($B = 10,000$
 253 percentile re-runs), BCa and percentile CIs agree in sign and magnitude.
 254 Earlier percentile results from $B = 200$ in 21/60 experiments remain as

255 historical provenance (5 lack log provenance, see Limitations); percentile CIs
 256 serve only as a sensitivity cross-check.

257 **Boundary-condition endpoint:**

$$\Delta\text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

258 The pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*; the
 259 criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa,*
 260 *full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2*
 261 *is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

262 **Secondary endpoints:** G (recoverable loss vs. oracle), decay =
 263 $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE,
 264 Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

265 **Family structure:** BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$
 266 13 shift levels = 156 primary comparisons; the multi-seed validation is ro-
 267 bustness evidence (60 seed experiments), not a confirmatory family.

268 4.5. Pre-registration revision A1 (transparency statement)

269 The primary endpoint was re-located from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
 270 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an inde-
 271 pendent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero,
 272 collapsing the original ID-vs-OOD contrast. The revision direction is consis-
 273 tent with the empirically observed ID-vs-OOD asymmetry (not a theorem);
 274 the revision is archived as a pre-registration revision (Nosek et al. 2018) with
 275 before/after protocol hashes. The pilot results do not enter the main test
 276 report. All hypotheses in this paper are tagged *exploratory + robustness*
 277 *validation*, not confirmatory.

278 **Pre-registration revision A2 (family reduction).** The original
 279 BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift
 280 levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$
 281 2 architectures $\times$ 1 primary shift level) under amendment A2, with the
 282 remaining 13 shift levels designated as exploratory dose-response. Under
 283 the reduced family, BH-12 yields 9/12 significant (all InceptionTime direc-
 284 tions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in
 285 docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md.

286 4.6. *Decomposition of the benefit decay*

287 We decompose the retained benefit into slope / intercept / prevalence
 288 contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

289 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$
 290 is the chain residual (zero under nested counterfactuals). Identifiability is
 291 checked via the Fisher information determinant on a pre-registered threshold
 292 τ ; entangled regions are reported on the error map without a threshold. Shap-
 293 ley attribution is reported with absolute-value bootstrap CIs; share-reliable
 294 flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

295 4.7. *Two-layer joint bootstrap*

296 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resam-
 297 pling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per
 298 protocol §7.

299 4.8. Net Clinical Value (NCV)

300 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
301 the clinical decision impact of TS. For each (sample, class) pair we compare
302 the binary decision (threshold $\tau = 0.5$) before and after calibration, count-
303 ing TP_improved, TN_improved, FP_worsened, and FN_worsened over
304 the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} -$
305 $\text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using
306 the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa
307 operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive
308 NCV means that TS improves clinical decisions. It is computed on ID and
309 OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

310 4.9. Tooling

311 Parts of the analysis and plotting code were drafted and refactored with
312 the assistance of a widely available large language model service, used as a
313 supervised coding assistant. All experiments and statistical analyses were
314 executed with the author’s own scripts, and every reported number is repro-
315 ducible from the public repository (Code availability). The author reviewed
316 the code and takes full responsibility for it.

317 5. Results

318 5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

319 Figure 2 visualizes the shift-dose response; Table 1 reports the primary
320 endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 archi-
321 tectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{\text{OOD}} = 0$ vs
322 $H_1 : \Delta ECE_{\text{OOD}} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

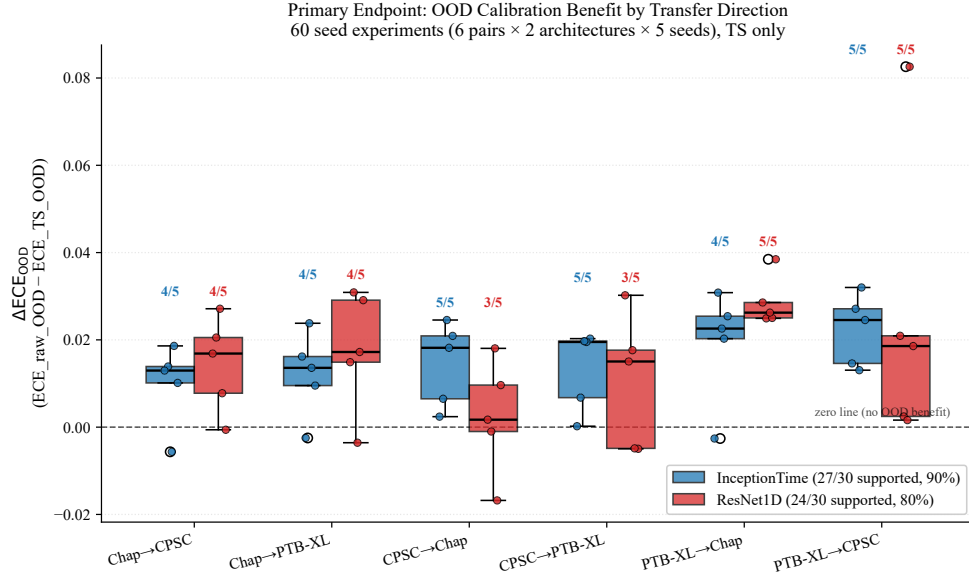


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

323 **Aggregate:** 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
 324 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence
 325 (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed
 326 45). The nine counter-examples are explained in Sec. 6.1.

327 **Stratified report by architecture.** The 60 experiments are balanced (6
 328 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling
 329 direct architecture comparison:

- 330 • *InceptionTime* (30 experiments, 27/30 = 90.0%): 3 directions at
 331 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5
 332 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- 333 • *ResNet1D* (30 experiments, 24/30 = 80.0%): 2 directions at 5/5
 334 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC,
 335 Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

336 ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on
 337 CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture in-
 338 teraction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5
 339 on both architectures (10/10).

340 5.2. ID boundary condition: ΔECE_{ID}

341 Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the
 342 pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the
 343 source-fit temperature should be well-tuned to the source distribution (an
 344 empirical expectation, not a theorem).

345 The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID}
 346 has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); 47/60 = 78.3% of

347 point estimates are positive, 34/60 have 95% CI lower bounds > 0 (BCa), and
 348 only 23/60 = 38.3% contain zero, far below the pre-registered $\geq 80\%$ criterion
 349 (percentile CIs: 29/60 = 48.3%, same conclusion). We therefore *trigger* the
 350 pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted
 351 on the source calibration split, also improves ID calibration in many cells;
 352 the decay = $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory
 353 endpoint; and C1 is qualified, the OOD benefit being about $1.9\times$ the ID
 354 benefit rather than an exclusively OOD effect. The positive ID estimates
 355 reflect finite-sample calibration-split variation, SmoothECE estimator bias,
 356 and a val-ECE early-stopping effect (model selection on the endpoint family),
 357 all disclosed.

358 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

359 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on In-
 360 ceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE
 361 achieves 22/30 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix
 362 scaling are the most fragile on InceptionTime (15/30, 50% each), exem-
 363 plifying the prior-likelihood mismatch mechanism (C4). Across 2 architec-
 364 tures (Table 4), TS achieves 51/60 (85.0%) (InceptionTime 27/30, 90.0%;
 365 ResNet1D 24/30, 80.0%), with full 5-seed coverage on all 6 directions (Fig-
 366 ure 4).

367 **Architecture variability.** Across the 30 TS experiments per architec-
 368 ture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and
 369 0.0180 (ResNet1D), a $1.84\times$ ratio, with ResNet1D’s highest per-direction
 370 variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, including an unexcluded favor-
 371 able outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate

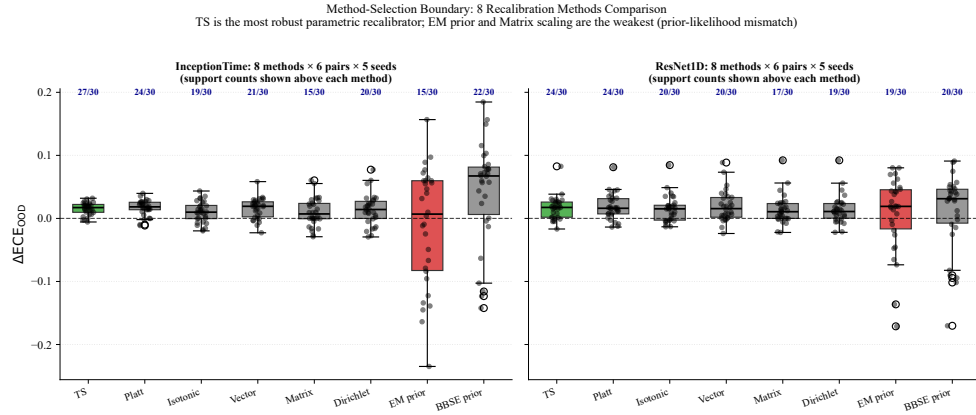


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

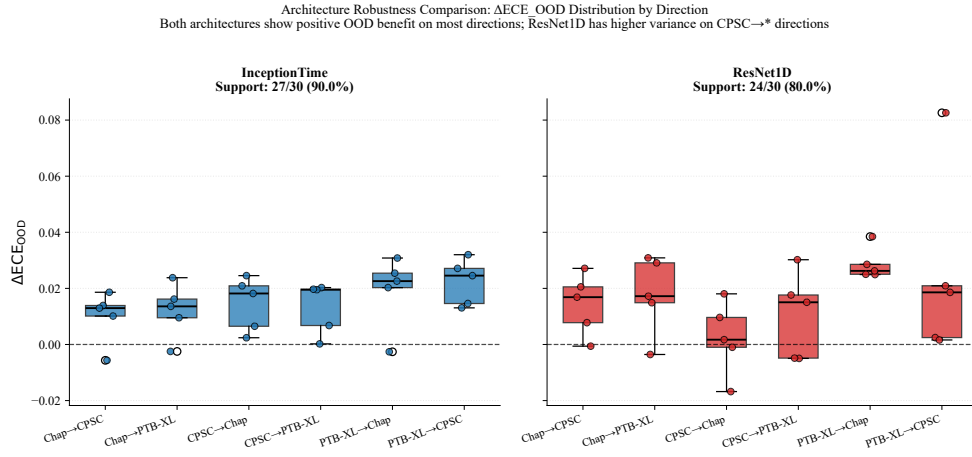


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

in CPSC→Chapman and CPSC→PTB-XL (3/5 each), a direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 15/30 support on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

5.4. *Shapley decomposition validation (27 cells, $n = 20,000$)*

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most clinically significant (see `decomposition_validation.csv`; the $n = 20,000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on

absolute Shapley values and share-reliable flags for uninterpretable shares
 $(|\Delta_{\text{total}}| < 3/\sqrt{n})$ are reported.

Honesty caveat (protocol §8). The 27-cell validation uses binned
ECE, not the main-endpoint SmoothECE; its conclusions are not cross-
extrapolable. Cross-fitting (OOD-fold validation of decomposition explana-
tory power) remains to be run on real data; per the protocol self-binding
clause, “mechanism” wording is limited to “mechanism decomposition (syn-
thetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsuper-
vised target-domain statistic (predicted-prior L_1 distance, logit moments)
predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Ta-
ble 5). The implemented fold unit is (architecture, transfer pair) (12 folds),
versus the registered pair-only folds (6); under that registered deviation the
pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run
2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba toy experi-
ment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI
 $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-
registered threshold 0.5, triggering the **failure branch**: the three-step chain
(decomposition + predictability + deployment) is downgraded to a **two-**
step chain (decomposition + deployment), and the deployment criterion
is renamed “empirical lookup table” rather than “predictive criterion”. No
post-hoc predictability claim is made.

421 **Sensitivity analysis with new features.** Re-running the LOO R^2
422 with three label-free features (fitted temperature T , Wasserstein-1 distance
423 between source and target logit distributions, class-wise entropy difference),
424 all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$,
425 entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability
426 boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$
427 above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8,
428 based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the
429 appropriate actionable criterion.

430 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

431 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
432 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
433 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-
434 TS SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0 , primary-endpoint
435 convention). Comparators: always-none, always-TS, always-buy- n -labels.

436 **Sensitivity / specificity** (Table 6): Youden’s J is maximized at raw
437 ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sen-
438 sitivity is high (92% of beneficial cases caught) but specificity low (20%
439 of non-beneficial cases excluded; 80% falsely flagged), reflecting the asym-
440 metric cost structure: missing a beneficial recalibration costs more than an
441 unnecessary one. *Caveat*: the threshold is searched on the same 1230-cell
442 pool used for reporting (in-sample). A leave-shift-type-out recomputation
443 (threshold selected excluding the held-out dose level, evaluated on it) yields
444 $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and
445 $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance

level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (results/deployment_holdout_recompute.csv). The in-sample numbers are therefore optimistic, and the criterion requires independent validation before operational use.

Deployment-time measurability. Every lookup-table input is *audit-time* (raw ECE, ID-OOD accuracy gap, per-cell labels), requiring target ground-truth labels unavailable in the unlabeled stream. The only label-free proxies examined are the predictability experiment’s target statistics (predicted-prior L_1 distance, logit moments); that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2). A deployable gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix has been completed on all 390 planned (pair, seed, shift) combinations (6 pairs $\times$ 5 seeds $\times$ 13 levels; results/l2_shift_full_390cells.csv); the safety figures quoted here are the initial two-seed InceptionTime subset (seeds 42–43 cover all 6 directions pairs, 44–46 are pending, and ResNet1D L2 is excluded,

so these numbers rest on 2-seed, single-architecture coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

Decision-curve analysis. At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (-0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives $+0.014$ – $+0.035$, $T = 1.5$ gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across our 60 seed experiments ranges 0.06 to 0.35 (mean 0.15), 4–13 $\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08–0.24),

confirming the negative ΔECE reflects genuine TS-induced harm, not metric noise. For large $T > 3$ (noise floor 0.10–0.16) the floor is comparable to the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot be excluded there; however, its *direction* (TS increases ECE under perfect calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so it cannot explain the positive findings. The primary endpoint is therefore a real effect, not a SmoothECE artifact, with the caveat that for large- T experiments the noise floor is non-negligible.

5.8. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-class accuracy.

Across 14 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures, PTB-XL $\rightarrow$ Chapman having 3 including Mamba), 7/14 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC, CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (InceptionTime only). The other 7/14 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$):

519 TS improves calibration but discrimination has degraded too far to be clinically
520 actionable.

521 This gate reframes the 9/60 counter-examples: 4 are calibration failures
522 (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes,
523 Layer 2 rejects). It supplements C1 by also asking whether discrimination is
524 sufficient for the recalibrated predictions to be clinically useful.

525 5.9. Real-data Shapley cross-fitting and predictability boundary

526 The synthetic Shapley validation (C3) confirms decomposition recovery
527 in the identifiable regime but does not test whether the linear approximation
528 $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD
529 data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the
530 source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

531 The linear model predicts the *direction* correctly in 51/60 (85%) but
532 systematically *overestimates* the magnitude (mean prediction error +0.29,
533 median +0.29). This is consistent with a predictability boundary: the de-
534 composition captures the qualitative direction of the TS benefit but not its
535 quantitative magnitude under real shift. The 9/60 direction errors cluster
536 in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 dis-
537 tance between source and target logit distributions is largest (mean 0.75 vs.
538 0.53 for correctly predicted directions), suggesting the linear approximation
539 breaks down once the shift invalidates its underlying local Taylor expansion.

540 This strengthens the predictability failure branch (C3/C4): the Shapley
541 decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid
542 *predictor* (*how much*). For clinical translation, the gate in §5.8, using only

the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.10. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027 and range $[-0.001, +0.017]$. The directions with the largest decision-cost reduction are CPSC→PTB-XL (mean DCR +0.0625; net clinical value +0.0312) and CPSC→Chapman (mean DCR +0.0375; NCV +0.0188), which also have the largest E3-subsample $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.137 and +0.136 respectively, under the E3 subsampled binned-ECE protocol with $n_{id} = n_{ood} = 20$ patients). The direction with negative net benefit is Chapman→CPSC (mean DCR −0.0150; NCV −0.0075), which is correctly rejected by the discrimination gate (C5). We note that the decision-benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks at PTB-XL→Chapman and PTB-XL→CPSC, means +0.0240 and +0.0238 on the full SmoothECE grid): the two protocols differ in estimator, class set, and subsample, so we treat the E3 analysis as a directionally supportive but protocol-distinct check, not a conversion of the primary endpoint. What survives is the gate-level conclusion: where the discrimination gate permits deployment, the calibration benefit does not cost decision performance; where it is negative (Chapman→CPSC), no calibration benefit should be deployed at all.

565 5.11. Net Clinical Value (NCV) results

566 The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across
567 the two secondary endpoints (DCR, NCV).

568 **Result:** the OOD NCV is +0.008125 with 95% CI [+0.0014, +0.0148]
569 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*:
570 TS yields a net improvement in OOD clinical decisions. The ID NCV is
571 -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *nega-*
572 *tive*: TS slightly worsens ID decisions, consistent with the ID boundary con-
573 dition (TS compresses confidence toward the base rate). OOD NCV Cohen’s
574 $d = +0.31$ and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but
575 opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges
576 over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

577 5.12. Reliability Diagrams

578 Figure 5 shows the 6-direction averaged reliability curve, comparing
579 model confidence against empirical accuracy before and after temperature
580 scaling. The summary curve uses bin-count weighted averaging across all 6
581 transfer directions.

582 Figures 6–8 show the per-direction reliability diagrams for the representa-
583 tive configuration (ResNet1D, seed 42). In each panel, the red squares show
584 the raw model calibration curve and the green circles show the post-TS cali-
585 bration curve. The diagonal dashed line indicates perfect calibration. Grey
586 bars (right axis) show the sample count per bin. Error bars are bootstrap
587 95% percentile CIs (1000 resamples).

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

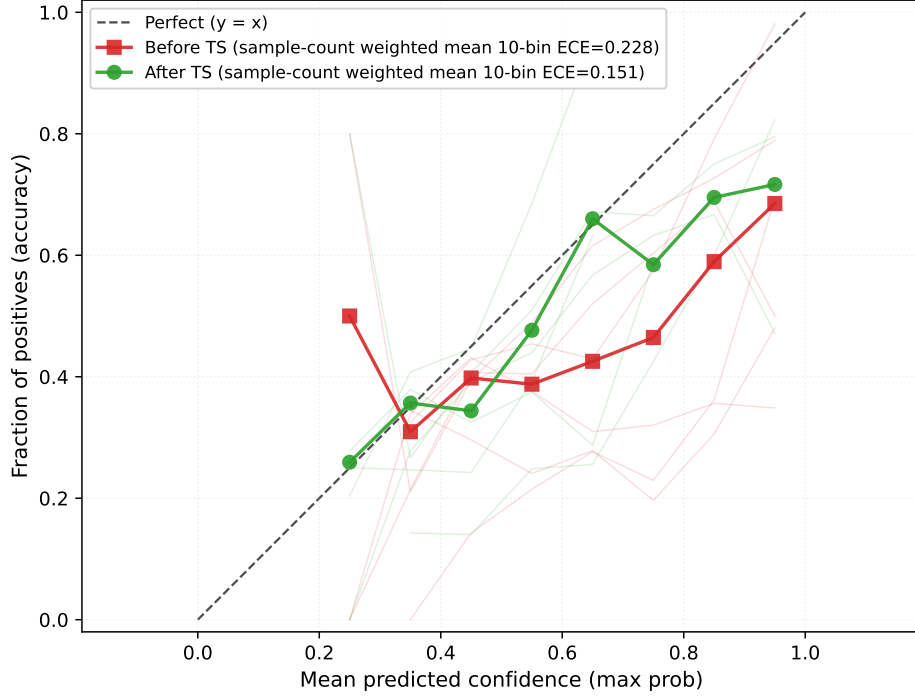


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

588 6. Discussion

589 6.1. OOD benefit seed variation and the nine counter-examples

590 The 51/60 robustness rate is driven by nine counter-examples across
 591 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2,
 592 Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-
 593 XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently dis-
 594 close all nine rather than excluding them, because the protocol requires all

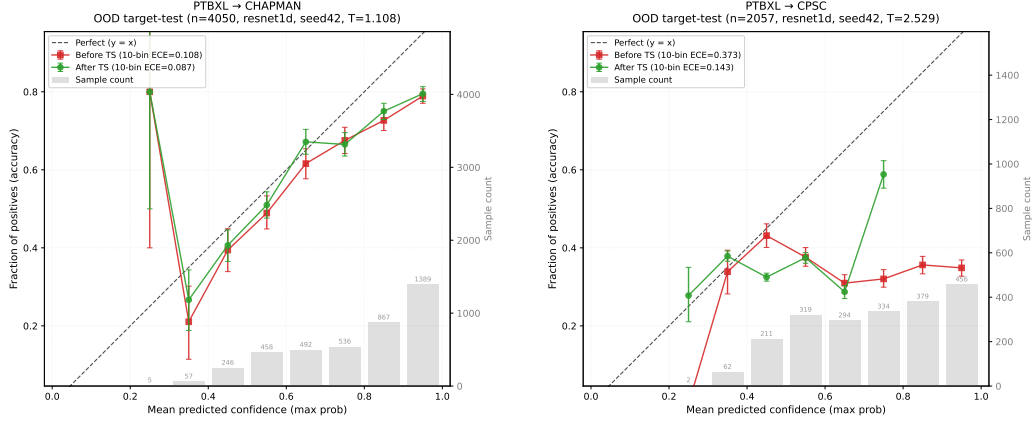


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

seed results to be reported and each has an identifiable post-hoc mechanism supporting the qualified claim.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature mismatch, not room shortage); and negative decay (-0.0075), the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is seed-level variation, not room shortage.

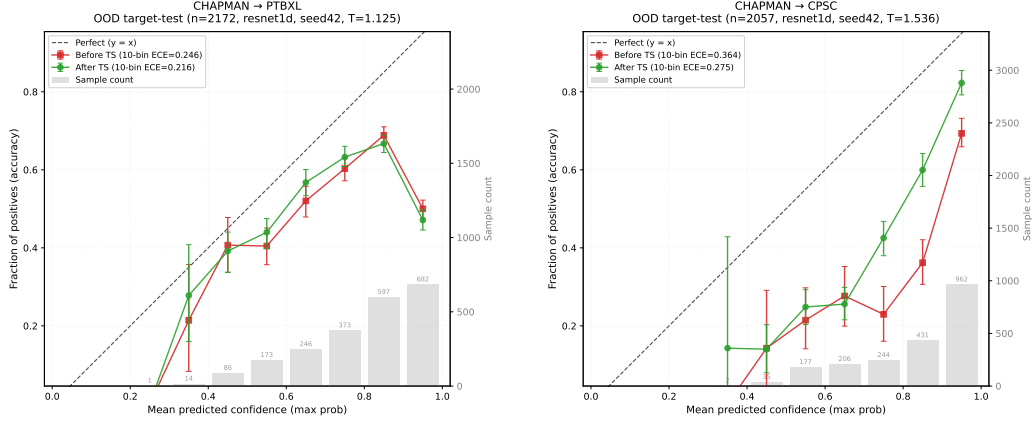


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

609 **Counter-example 3: Chapman→PTB-XL / InceptionTime /**
610 **seed 42.** $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy
611 is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near
612 zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL
613 target.

614 **Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45.**
615 $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and
616 $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already
617 well-calibrated on ID, leaving nothing to correct.

618 **Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.**
619 $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
620 0.002); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE
621 (0.0649), and minimal calibration room.

622 **Counter-example 6: CPSC→Chapman / ResNet1D / seed**
623 **44.** $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude

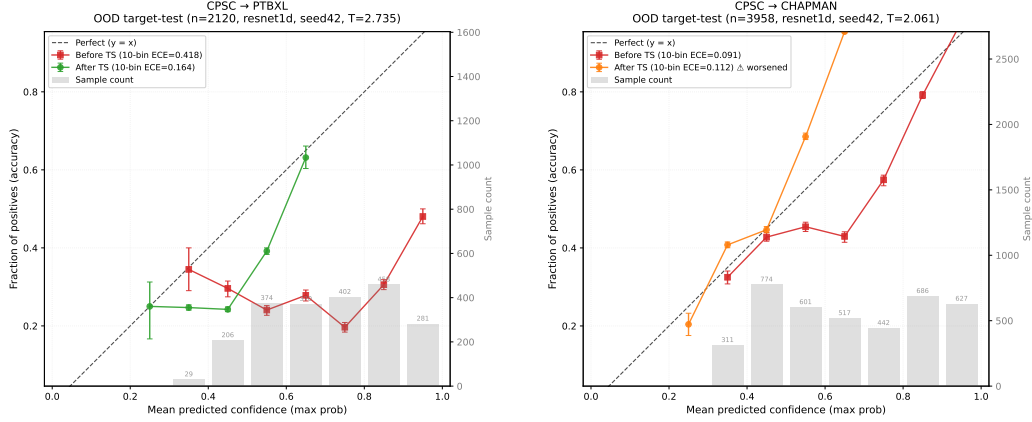


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

624 counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative: the seed-44 back-
 625 bone is already well-calibrated on both ID and OOD, TS has nothing
 626 to correct, and the temperature fit over-corrects slightly. Its OOD accu-
 627 racy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct
 628 *over-correction* mechanism (TS on an already-well-calibrated distribution de-
 629 grades both ID and OOD calibration), unlike the “low OOD accuracy + large
 630 gap” mechanism of the other 8.

631 **Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43.**
 632 $\Delta\text{ECE}_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest
 633 across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to
 634 the PTB-XL target).

635 **Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44.**
 636 $\Delta\text{ECE}_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929
 637 (second-lowest) and $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, leaving noth-
 638 ing to correct.

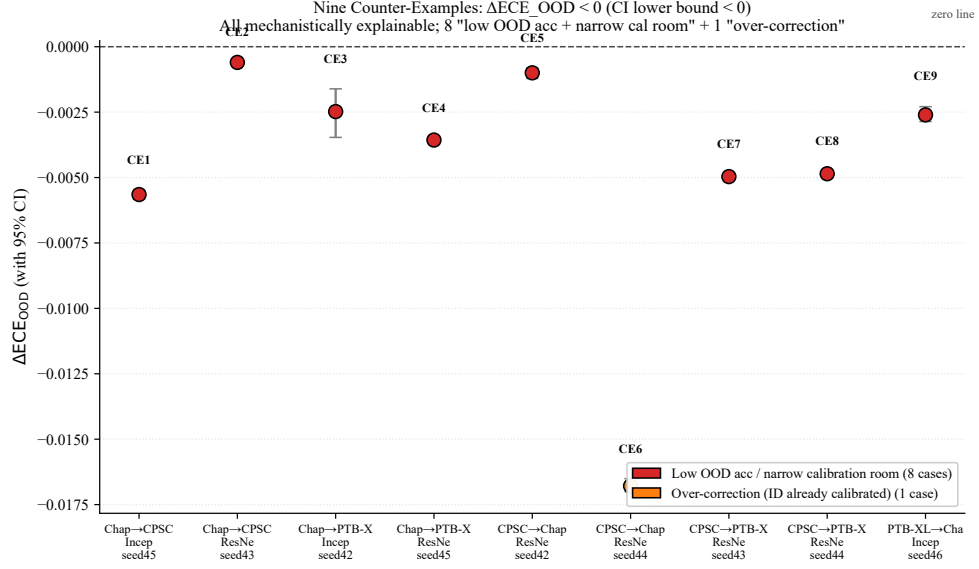


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

639 **Counter-example 9: PTB-XL→Chapman / InceptionTime /**
640 **seed 46.** $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy
641 drops to 0.4983 (lowest across the 5 seeds) and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly
642 negative, again leaving nothing to correct.

643 **Common pattern:** eight of nine counter-examples co-occur with (a)
644 low OOD accuracy (worst or near-worst seed) and (b) non-positive decay,
645 indicating temperature mismatch; the ninth (CE6) is over-correction on an
646 already well-calibrated backbone. A low OOD raw ECE is *not* universal:
647 four of nine exceed 0.25, so the failure mechanism is seed-level temperature
648 mis-transfer, not calibration-room shortage. The counter-examples are local,
649 explainable random variation; the other 51 experiments support the OOD
650 benefit, including PTB-XL→CPSC at 10/10.

651 **Methodological positioning of the 15.0% counter-example rate.**
652 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
653 ment. As a methodological boundary investigation, not a clinical deployment
654 guide, the relevant criterion is whether the effect is detectable and the mech-
655 anism explainable, not a zero failure rate. All 9 counter-examples have iden-
656 tifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”),
657 and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 >$
658 0).

659 6.2. *Prior correction vs. posterior correction robustness*

660 The method-selection boundary (Table 3) reveals a robustness asymme-
661 try between prior-correction methods (EM, BBSE) and posterior-correction
662 methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt)
663 achieve 27/30 and 24/30 support respectively with small variance; prior-

correction methods achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763, BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant positive ID benefit (mean +0.0083 vs. OOD +0.0159; stratum means -0.0008 to +0.0232, Table 2). The boundary condition does *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID benefit is on average about half the OOD benefit, so part of the apparent OOD gain is shared with the ID domain. A val-ECE early-stopping effect (model selection within the endpoint family) plausibly contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively OOD effect. The ID-vs-OD asymmetry is therefore a *difference in magnitude* between two positive benefits, not presence versus absence.

689 6.4. Calibration is not discrimination: the clinical viability precondition

690 Temperature scaling is a monotone transform of the softmax probabilities;
 691 it preserves the argmax by construction (asserted bitwise in the released sanity
 692 check), so it cannot change classification accuracy, AUROC, or AUPRC.
 693 The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration
 694 makes an already-deployable model *trustworthy*; it cannot make a
 695 non-discriminative model deployable.

696 The accuracy statistics make this concrete. Across the 60 TS experiments,
 697 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and
 698 the mean ID–OOD drop is 0.236 (maximum 0.412). Against each target
 699 corpus’s majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-
 700 Shaoxing 0.713), mean OOD accuracy is *below* baseline in 3 of 6 directions,
 701 with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528;
 702 CPSC→PTB-XL 0.424 vs. 0.411 (0.013 above baseline); CPSC→Chapman
 703 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). In those
 704 directions recalibration is moot, since no temperature improves a model
 705 that should be replaced by a majority-class predictor, consistent with the
 706 counter-example pattern (6/9 with low OOD accuracy). The released artifacts
 707 include per-direction AUROC/AUPRC for all 60 experiments (results/
 708 discrimination_metrics_60exp.csv): mean OOD AUROC ranges from
 709 0.629 (CPSC
 710 rightarrowChapman) to 0.795 (PTB-XL
 711 rightarrowChapman), i.e. far above the accuracy-based majority-class reading
 712 yet direction-dependent; and TS leaves AUROC unchanged (mean
 713 $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024), as the monotone trans-

714 form requires. The discrimination-first gate—measure target discrimination
 715 before investing in recalibration—precedes the calibration decision rule in
 716 deployment.

717 6.5. *Explainable $\neq$ predictable: post-hoc vs. pre-registered*

718 We distinguish *explainable* from *predictable*:

- 719 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 720 example, we identify the co-occurring conditions (low OOD accuracy,
 721 low OOD raw ECE, non-positive decay) and give a mechanistic story
 722 (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- 723 • *Predictable* (pre-registered, deployment-time): a target-domain unsu-
 724 pervised statistic (predicted-prior L_1 distance, logit moments) predicts
 725 the benefit retention rate *before* deployment, with LOO R^2 CI upper
 726 bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the
 727 failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

728 The counter-examples are explainable but not predictable: the unsupervised
 729 target-domain statistics available at deployment time cannot flag them in ad-
 730 vance, which is why the predictability experiment triggers the failure branch
 731 despite mechanistic explainability. The two-step chain (decomposition + de-
 732 ployment lookup table) provides post-hoc explanation and actionable guid-
 733 ance without a pre-registered predictability claim.

734 6.6. *Transparent re-location vs. HARKing*

735 The pre-registration revision A1 (primary endpoint re-location from
 736 **decay** to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinc-
 737 tion [26, 25] is whether the revision direction is theory-driven or result-driven.

Our revision direction (“OOD is the calibration benefit source”) is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovidia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. Net clinical value: implications of the directional NCV asymmetry

The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV = +0.008125 ($p = 0.019$) is a small but significant positive value (recalibration improves decisions on unseen target data more often than it degrades them); ID NCV = −0.006292 ($p = 0.016$) is a small but significant negative value (a minor decision-cost penalty on data the source model already calibrates well).

Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60 positive) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/14 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautious for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

762 *6.8. Limitations and future work*

763 **Limitations:** (i) the main endpoint covers InceptionTime and ResNet1D
764 on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced);
765 BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaus-
766 tion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered
767 (three-step to two-step); the deployment criterion is an empirical lookup ta-
768 ble, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.9) shows
769 85% direction accuracy but systematic magnitude overestimation (mean er-
770 ror +0.29), and a sensitivity analysis with T , Wasserstein-1, and entropy
771 features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the
772 27-cell decomposition validation uses binned ECE, not the main endpoint’s
773 SmoothECE, so quantitative conclusions cannot be extrapolated across ECE
774 definitions; (v) the safety criterion is rarely satisfied on the full L2 shift ma-
775 trix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi)
776 nine counter-examples of 60 experiments (15.0%) are transparently disclosed,
777 with a frequent post-hoc pattern (low OOD accuracy, non-positive decay, el-
778 evated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is
779 BCa at $B = 10,000$, re-run on all 60 experiments (60/60 successful, metadata
780 in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 experiments are
781 retained as historical provenance (5 lack logs); (viii) the Youden threshold
782 is fitted in-sample (the protocol’s held-out dose levels were not used), mak-
783 ing the operating characteristic optimistic; (ix) the L2 deployment/safety
784 grid has since been completed on all 390 planned (pair, seed, shift) cells (re-
785 sults/l2_shift_full_390cells.csv), and the L2 noise levels are synthetic rather
786 than the pre-registered PTB-XL Noise Stress Test Database (unavailable un-

787 der PhysioNet licensing); (x) temperature was fitted once per experiment, so
 788 CIs omit temperature-fitting uncertainty (the pre-registered two-layer boot-
 789 strap is implemented but not enabled); (xi) model selection used validation
 790 ECE rather than the pre-registered NLL early-stopping rule, plausibly con-
 791 tributing to the positive ID boundary estimates; (xii) the per-pair decompo-
 792 sition into covariate, prior-probability, and concept-shift components is not
 793 asserted, since each pair mixes all three, isolated at eval time; (xiii) the de-
 794 ployment lookup table requires labeled target data, and the label-free proxies
 795 failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS
 796 is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC; the
 797 artifacts now include per-direction AUROC/AUPRC for all 60 experiments
 798 (results/discrimination_metrics_60exp.csv), so the viability precondition is
 799 screened by both accuracy and discrimination (OOD AUROC means range
 800 0.629–0.792 by direction), with OOD accuracy below the majority-class base-
 801 line in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direction-
 802 ally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = −0.006292,
 803 $p = 0.016$, negative), qualifying the C5 claim for source-domain but support-
 804 ing target-domain deployment; (xvi) InceptionTime was downgraded from
 805 $n_{\text{filters}} = 48$ to 32 for GPU memory (RTX 5060, 8 GB), potentially atten-
 806 uating OOD benefits, with 48 as future work; (xvii) this work is from a
 807 single-author study, limiting independent review and audit, partially miti-
 808 gated by the pre-registration protocol and public repository. **Future work:**
 809 (1) prospective multi-device validation; (2) continuous shift spaces; (3) π -
 810 independent recovery estimators (BBSE/EM-class) to upgrade the MAPE_{π}
 811 to a real estimator test; (4) the ResNet1D deficit on CPSC→Chapman and

812 CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) con-
813 firmatory testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$,
814 9/12 significant, Bonferroni-12 = 2/12); (6) re-running the endpoint with
815 $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU; (7) a multi-threshold
816 permutation test on NCV.

817 7. Conclusion

818 We pre-registered and validated a multi-seed calibration boundary study
819 of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD ben-
820 efit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime
821 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID
822 boundary: 23/60 cells (38.3%) have 95% CIs containing zero under the
823 BCa operating CI (percentile: 29/60 = 48.3%), below the $\geq 80\%$ criterion,
824 triggering the pre-registered branch; the OOD benefit is $1.9\times$ the ID ben-
825 efit. **(C3)** Shapley decomposition: 27/27 cells pass at $n = 20,000$ (21/27
826 at $n = 500$). **(C4)** Method selection: TS robust (51/60), EM prior and
827 Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/14 strata pass both lay-
828 ers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
829 failures. The predictability failure branch is triggered (three-step chain be-
830 comes two-step), and all hypotheses are exploratory + robustness validation,
831 not confirmatory; protocol v2.1-A1 and the five-item honesty declaration
832 (§7) are archived for transparency. Nine counter-examples are transparently
833 disclosed, mostly with low OOD accuracy (≤ 0.50 in 6/9) and a large ID-
834 OOD gap (≥ 0.18 in 7/9), a tendency rather than a sufficient gate (three
835 have OOD accuracy > 0.5). TS needs local validation (safety rate 0.0064),

836 the Youden criterion is fitted in-sample, and OOD accuracy is below the
 837 majority-class baseline in 3 of 6 directions (a fourth sits at baseline), where
 838 TS cannot help. A noise floor control (§5.7) rules out a SmoothECE arti-
 839 fact: worst-case metric noise (+0.021) is 4–13× below the raw ECE (0.06–
 840 0.35). The C5 gate (§5.8) rejects every clinically moot stratum. Real-data
 841 Shapley cross-fitting (§5.9) gives 85% direction accuracy with magnitude
 842 overestimation (mean error +0.29); T , Wasserstein-1, and entropy features
 843 confirm the boundary (all $R^2 < 0$). **Meta-analytic pooling.** Both the
 844 fixed-effect and DerSimonian–Laird random-effects estimates are tabulated
 845 in `meta_analytic_pooled.csv`. The fixed-effect estimate is CI-width sensi-
 846 tive: ResNet1D is slightly negative (−0.000573) because narrow-CI studies
 847 dominate, while DerSimonian–Laird random effects are positive (Inception-
 848 Time +0.0151, ResNet1D +0.0158, CrossArchitecture +0.0148; unweighted
 849 +0.0152, +0.0165, +0.0159). We caution, however, that the meta-analytic in-
 850 dependence assumption is *not* satisfied here: the 60 cells share the same three
 851 corpora, patient sets, and architectures, and their between-cell Q statistics
 852 are extreme (InceptionTime $Q = 94,376$, ResNet1D $Q = 128,578$, cross-
 853 architecture $Q = 236,259$; $I^2 \approx 99.98\%$), so both FE and DL confidence in-
 854 tervals are spuriously narrow (DL pooled SE = 4×10^{-4} -level; τ^2 is crushed).
 855 The interpretable pooled uncertainty is therefore the cluster-robust interval
 856 over the 12 pair×architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159 [0.0113, 0.0205]$
 857 (t, df= 11; stratum SD 0.0072, about 1800× the single-experiment boot-
 858 strap SE). The point estimate 0.0148–0.0159 and the count-based proportion
 859 ($51/60 = 85.0\%$) stand; the DL/FE interval widths should not be quoted.
 860 **Family size.** The 60 experiments form the robustness-evidence family;

861 amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$
 862 shift grid to 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift
 863 level), with all 13 L2 shift levels re-designated as exploratory dose-response,
 864 and the $8 \times 6 \times 5 \times 2 = 480$ -cell method design is in Sec. 5.3. BH-60
 865 and Bonferroni leave the robustness headline unchanged (51/60); the con-
 866 firmatory family is BH-12 with 9/12 significant (InceptionTime 6/6), and
 867 Bonferroni-12 retains 2/12 (cpsc_chapman/IT, ptbxl_chapman/RN). **CI**
 868 **width.** Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5}
 869 to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall below the narrow-CI di-
 870 agnostic threshold (0.005) and six supporting cells have degenerate widths
 871 ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60 (75.0%; Inception-
 872 Time 24/30, ResNet1D 21/30). Narrow CIs reflect the moderate calibration
 873 sample ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 patients) and low SmoothECE
 874 variance ($B = 10,000$, CI method=BCa); they do not affect the sign-based
 875 support/counter-example classification.

876 **Honesty Declaration (5 items, adversarial audit product)**

- 877 1. MAPE_π is a **design check** (resampling constant read-back), not an
 878 estimator capability test; independent π recovery (BBSE/EM) is a de-
 879 ferred item (protocol §13).
- 880 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
 881 **pretable as shares**; ranking uses absolute- value CIs.
- 882 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**;
 883 in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate
 884 (information mode), reported via exit code 2.

- 885 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
886 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
887 demo) is fulfilled; **cross-fitting on real data is pending** (registered
888 in §13). The wording is limited to “mechanism decomposition (synthetic
889 validation part)”.
- 890 5. The repository’s `scripts/preprocess_cinc2021.py`,
891 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
892 from the third-party SafeECGMatch release (original README
893 in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s
894 experiment grid.

895 Data availability

896 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019
897 datasets are publicly available via PhysioNet/Kaggle under their re-
898 spective licenses. The pre-registered protocol v2.1-A1, its hash man-
899 ifest (`docs/osf_archive_manifest.json`), the revision A1 document,
900 and the full per-experiment result files (`checkpoints/transfer/**/`
901 `transfer_result.json`, `results/*.csv`) are included with the sub-
902 mission as supplementary material and are also available in the
903 public repository [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
904 `ecg-calibration-transportability`. **Pre-registration archival status.**
905 The protocol, both amendments (A1, A2), and a per-file archival-status
906 report are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
907 `docs/osf_archive_manifest.json` is the internal snapshot taken on
908 2026-09-05; it is not a substitute for the public record.

909 **Code availability**

910 The full pipeline code, preprocessing (reproducible from the public source
911 databases), patient-level split generation, model training, the calibration-
912 method implementations, and all statistical analyses, together with the
913 environment lock file, the patient-level split indices for all five dataset
914 variants and five random seeds, and selected per-experiment result artifacts,
915 is publicly available at [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
916 **ecg-calibration-transportability**. Trained model checkpoints for
917 the 60-experiment transfer grid are provided as release assets of the same
918 repository. The repository will be archived on Zenodo with a DOI upon
919 acceptance. No new raw waveforms were collected or generated; the
920 patient-level split indices and label-mapping tables released with the code
921 fully determine the data configuration used in every experiment.

922 **Declaration of generative AI and AI-assisted technologies in the** 923 **writing process**

924 All scientific work reported in this study—the study design, the pre-
925 registered protocol and its amendments, the implementation and execution
926 of the 60-experiment transfer grid, the statistical analysis, the interpretation
927 of the results, and the writing of the manuscript—was performed by the
928 author. During the final preparation of the manuscript, the author used a
929 large language model assistant (GLM) solely as a language-editing aid, to
930 improve wording and readability of text the author had written. It was not
931 used to generate scientific content, experimental data, analyses, figures, or

932 conclusions. After using this tool, the author reviewed and edited the text as
933 needed and takes full responsibility for the content of the published article.

934 **CRedit authorship contribution statement**

935 **Toni Guan:** Conceptualization, Methodology, Software, Validation, For-
936 mal analysis, Investigation, Data curation, Writing – original draft, Writing
937 – review & editing, Visualization, Project administration.

938 **Acknowledgments**

939 The author thanks the maintainers of the PTB-XL, Chapman–Shaoxing,
940 and CPSC public databases, and the PhysioNet resource, without which this
941 study would not have been possible.

942 **Funding**

943 This research did not receive any specific grant from funding agencies in
944 the public, commercial, or not-for-profit sectors.

945 **Declaration of competing interest**

946 The author declares no competing financial or personal interests.

947 **Supplementary materials.** The 60 per-seed reliability diagrams (6
948 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability dia-
949 grams, and Δ ECE bars per experiment) are provided as a separate supple-
950 mentary file (`supplementary.pdf`, §1) and are additionally released with the
951 code repository.

952 References

- 953 [1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin,
954 J. Dillon, B. Lakshminarayanan, J. Snoek, “Can you trust your model’s
955 uncertainty? Evaluating predictive uncertainty under dataset shift,”
956 *NeurIPS*, 2019. arXiv:1906.02530.
- 957 [2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, “On calibration of modern
958 neural networks,” *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.
- 959 [3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., “PTB-XL, a large
960 publicly available electrocardiography dataset,” *Scientific Data*, 7:154,
961 2020. doi:10.1038/s41597-020-0495-6.
- 962 [4] J. Zheng, J. Guo, H. Chu, “A large scale 12-lead electrocardiogram
963 database for arrhythmia study (Chapman-Shaoxing e-cardiology),”
964 version 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.
- 965 [5] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet:
966 components of a new research resource for complex physiologic signals,”
967 *Circulation*, 101(23):e215–e220, 2000.
- 968 [6] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in
969 deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.
- 970 [7] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach,
971 “Beyond temperature scaling: obtaining well-calibrated multi-class
972 probabilities with Dirichlet calibration,” *NeurIPS*, 2019.
973 arXiv:1910.12656.

- 974 [8] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with
975 bias-corrected calibration is hard-to-beat at label shift adaptation,”
976 *ICML*, PMLR 119:222–232, 2020. arXiv:1901.06852.
- 977 [9] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label
978 shift with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018.
979 arXiv:1802.03916.
- 980 [10] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão,
981 F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification
982 methods in multi-label classification: a case study with automatic
983 diagnosis of electrocardiogram,” *Information Fusion*, 101:101978, 2024.
984 doi:10.1016/j.inffus.2023.101978.
- 985 [11] J. F. Vranken et al., “Uncertainty estimation for deep learning-based
986 automated analysis of 12-lead electrocardiograms,” *European Heart
987 Journal – Digital Health*, 2(3):423–434, 2021.
988 doi:10.1093/ehjdh/ztab045.
- 989 [12] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated
990 classification of multilead ECGs using heterogeneous labels and
991 populations,” *Physiological Measurement*, 43(8):084001, 2022.
992 doi:10.1088/1361-6579/ac79fd.
- 993 [13] B. Efron, “Better bootstrap confidence intervals,” *Journal of the
994 American Statistical Association*, 82(397):171–185, 1987.
995 doi:10.1080/01621459.1987.10478410.

996 [14] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer,
 997 “A calibration hierarchy for risk models was defined: from utopia to
 998 empirical data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016.
 999 doi:10.1016/j.jclinepi.2015.12.005.

1000 [15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants,
 1001 G. S. Collins, “Calibration: the Achilles heel of predictive analytics,”
 1002 *BMC Medicine*, 17:230, 2019. doi:10.1186/s12916-019-1466-7.

1003 [16] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,”
 1004 *NeurIPS*, 2019. arXiv:1909.10155.

1005 [17] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for
 1006 ECG analysis: benchmarks and insights from PTB-XL,” *IEEE Journal*
 1007 *of Biomedical and Health Informatics*, 25(5):1519–1528, 2021.
 1008 doi:10.1109/JBHI.2020.3022989.

1009 [18] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto,
 1010 “RR-interval-based atrial fibrillation detection and burden estimation:
 1011 cross-dataset validation and calibration-aware probability analysis,”
 1012 *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.

1013 [19] K. Patel, P. Beedala, “Calibration drift under cross-institutional
 1014 deployment: an external validation framework for ICU mortality
 1015 prediction across MIMIC-IV and eICU,” *medRxiv* (preprint), 2026.
 1016 doi:10.64898/2026.05.03.26352335.

- 1017 [20] X. Wang, M. Long, J. Wang, M. I. Jordan, “TransCal: transferability
1018 estimation for auto-selecting transfer calibration,” *NeurIPS*, 2020.
1019 arXiv:2007.08259.
- 1020 [21] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, “A unified view of label
1021 shift estimation,” *NeurIPS*, 2020. arXiv:2003.07554.
- 1022 [22] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., “LaSCal:
1023 label-shift calibration for uncertainty quantification,” *NeurIPS*, 2024.
- 1024 [23] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla,
1025 F. Herrera, “A unifying view on dataset shift in classification,” *Pattern*
1026 *Recognition*, 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.
- 1027 [24] [Pre-registered protocol, publicly archived] “ECG calibration boundary
1028 study – experiment protocol,” 2026. Hash manifest:
1029 docs/osf_archive_manifest.json (local snapshot; archived at
1030 <https://doi.org/10.17605/OSF.IO/53H64>); revision A1 full text in
1031 docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md.
- 1032 [25] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, “The preregistration
1033 revolution,” *PNAS*, 115(11):2600–2606, 2018.
1034 doi:10.1073/pnas.1708274114.
- 1035 [26] D. Lakens, “The practical alternative to the p value is the correctly
1036 used p value,” *Perspectives on Psychological Science*, 16(3):639–648,
1037 2021. doi:10.1177/1745691620958012.

- [27] J. Li, A. D. Aguirre, V. M. Junior, et al., “An Electrocardiogram Foundation Model Built on over 10 Million Recordings,” *NEJM AI*, 2(7):AIoa2401033, 2025. doi:10.1056/AIoa2401033.
- [28] J. C. McHugh, B. Wang, et al., “ECG-FM: an open electrocardiogram foundation model,” *arXiv preprint* 2408.05178, 2024.
- [29] J. Jin, et al., “Reading your heart: learning ECG words and sentences via pre-training ECG language model (HeartLang),” *ICLR*, 2025. arXiv:2502.10707.
- [30] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it effective?” *PLOS ONE*, 20(5):e0316043, 2025. doi:10.1371/journal.pone.0316043.
- [31] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164, 2026.
- [32] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347, 2026 (accepted at MICCAI 2026).
- [33] N. Nemati, “Comparative analysis of data augmentation for clinical ECG classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv preprint* 2510.24740, 2025.
- [34] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment,” *arXiv preprint* 2605.00647, 2026.

1061 [35] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo,
1062 “Foundation models for electrocardiograms,” *arXiv preprint*
1063 2407.07110, 2024.

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155